Project Design Phase
Proposed Solution Template

Date	03 April 2026
Team ID	SWTID-2026-8629
Project Name	Smart Library
Maximum Marks	2 Marks

1. Technical Architecture

The Smart Library System follows a **3-tier architecture** consisting of:

- **Presentation Layer (UI)** – Handles user interaction (Student & Librarian)
- **Application Layer (Backend)** – Processes business logic (book issue, return, management)
- **Data Layer (Database)** – Stores and retrieves book and user data

Architecture Description:

- Users (Students/Librarians) interact through a web interface
- Requests are sent to the backend server
- Backend processes operations like book search, issue, and return
- Backend communicates with the database to store/retrieve data
- External APIs may be used for authentication and notifications
- Data is stored securely in cloud/local databases

2. Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for students & librarians	HTML, CSS, JavaScript, React JS
2	Application Logic-1	Handles user registration & login	Node.js / Django / Spring Boot
3	Application Logic-2	Handles authentication & validation	Firebase Auth / JWT
4	Application Logic-3	Business logic for library management (Books, Issue, Return)	Node.js / Python / Java
5	Database	Stores books, users, issue records	MySQL / MongoDB
6	Cloud Database	Cloud-based storage	Firebase / AWS RDS
7	File Storage	Stores book images/documents	AWS S3 / Local Storage
8	External API-1	Email/Notification services	Gmail API / Twilio API
9	External API-2	Authentication services	Google OAuth
10	(Optional) Analytics	Library usage analysis	Python / BI Tools
11	Infrastructure	Deployment environment	AWS / Azure / Local Server

3. Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for development	React JS, Node.js, Django

S.No	Characteristics	Description	Technology
2	Security Implementations	Protects user and library data	HTTPS, JWT, SHA-256
3	Scalable Architecture	Supports increasing users & books	3-tier / Microservices
4	Availability	System accessible anytime	Cloud hosting, Load Balancer
5	Performance	Fast response & efficient system	Caching (Redis), CDN

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4. Conclusion

The chosen technology stack ensures:

- Efficient library system performance
- High security and reliable data handling
- Scalability for growing users and book records
- Easy maintenance and deployment

This architecture provides a strong foundation for building a robust and user-friendly **Smart Library Management System**.